Python Programming Practice Problems

Assignment-4

1. Write a Python function greet() that prints **"Welcome to Python Programming"**.
2. Write a function show_name(name) that takes a name as a parameter and prints a greeting.
3. Write a function square(n) that returns the **square of a number**.
4. Write a function cube(n) that returns the **cube of a number**.
5. Write a function add(a, b) that returns the **sum of two numbers**.
6. Write a function largest(a, b) that returns the **larger of two numbers**.
7. Write a function even_odd(n) that checks whether a number is **even or odd**.
8. Write a function factorial(n) that calculates the **factorial of a number**.
9. Write a function average(a, b, c) that returns the **average of three numbers**.
10. Write a function count_digits(n) that returns the **number of digits in a number**.
11. Write a function list_sum(lst) that returns the **sum of all elements in a list**.
12. Write a function largest_number(lst) that returns the **largest element in a list**.
13. Write a function smallest_number(lst) that returns the **smallest element in a list**.
14. Write a function count_even(lst) that counts the **number of even numbers in a list**.
15. Write a function reverse_list(lst) that returns the **reversed list**.
16. Write a Python program to **plot a simple line graph** showing numbers from 1 to 10.
17. Write a Python program to **plot a line graph of squares of numbers from 1 to 10**.
18. Write a Python program to **plot a bar chart** showing marks of five students.
19. Write a Python program to **plot a bar chart** showing sales of products (A, B, C, D).
20. Write a Python program to **plot a pie chart** showing percentage distribution of subjects (Math, Physics, Chemistry, Biology).
21. Write a Python program to **plot a line graph showing temperature changes over 7 days**.
22. Write a Python program to **plot a bar graph showing the number of students in different courses**.
23. Write a Python program to **create a pie chart showing daily activities (study, sleep, play, others)**.
24. Write a Python program to **plot a graph of x and x^2 values for numbers from 1 to 10**.
25. Write a Python program to **display a scatter plot of two lists of numbers**.
26. Write a function to **generate a list of numbers from 1 to 10** and plot them using a line graph.
27. Write a function that **calculates squares of numbers in a list** and visualize them using a bar chart.

28. Write a function that **calculates average marks of students** and visualize them using a bar graph.
29. Write a program that **plots two lines on the same graph** (e.g., numbers and their squares).
30. Write a Python program that **plots a histogram for a list of random numbers**.